Project Planning Phase

Product Backlog, Sprint Planning, Stories & Story Points

Date	7 March 2026
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Project Name	Heart Disease Analysis
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule & Estimation

The product backlog captures all user stories and tasks required to build the Heart Disease Analysis Tableau dashboard. Stories are organized into two sprints based on priority and dependency. Each story is assigned story points using the Fibonacci scale to estimate effort.

Sprint	Functional Requirement (Epic)	Story No.	User Story / Task	Story Points	Priority	Team Member
SPRINT 1 — Data Setup & Core Visualizations | Total Story Points: 9						
Sprint-1	Data Collection & DB Setup	USN-1	As a team, we can import Heart_new2.csv into MySQL, verify schema, and confirm all 18 columns are correctly mapped	3	High	Team Member
Sprint-1	Gender vs Heart Disease	USN-2	As a user, I can view a stacked bar chart showing heart disease count by gender (Female: 2604, Male: 1896)	2	High	Team Member
Sprint-1	Age vs Heart Disease	USN-3	As a user, I can view heart disease distribution across all 13 age categories segmented by gender	2	High	Team Member
Sprint-1	Smoking & Alcohol Impact	USN-4	As a user, I can view how smoking and alcohol combinations affect heart disease count in a bar chart	2	High	Team Member
SPRINT 2 — Advanced Visualizations, Dashboard & Publishing | Total Story Points: 12						
Sprint-2	BMI & Diabetic Analysis	USN-5	As a user, I can view a treemap of age groups and average BMI values segmented by diabetic status	2	Medium	Team Member
Sprint-2	Race-wise Distribution	USN-6	As a user, I can view a pie chart showing heart	2	Medium	Team Member

			disease prevalence across all 6 racial groups			
Sprint-2	Physical Activity & General Health	USN-7	As a user, I can view physical activity donut chart (3,231 active vs 1,269 inactive) and general health bubble chart	2	Medium	Team Member
Sprint-2	Stroke & Disease Co-occurrence	USN-8	As a user, I can view stroke vs diabetes scatter plot and stroke vs asthma/kidney/skin cancer bar chart	2	High	Team Member
Sprint-2	Dashboard & Story Assembly	USN-9	As a user, I can view all 10 sheets in a combined Tableau Dashboard and Story titled Heartbeat of Data	3	High	Team Member
Sprint-2	Publish to Tableau Public	USN-10	As a team, we can publish the final packaged workbook (.twbx) to Tableau Public for stakeholder access	1	High	Team Member

Project Tracker, Velocity & Burndown Chart

Sprint Summary Tracker

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	14	7 Days	4 March 2026	10 March 2026	14	10 March 2026
Sprint-2	12	7 Days	11 March 2026	17 March 2026	TBD	TBD

Velocity Calculation

Velocity measures the amount of work a team completes during a sprint, expressed in story points per day. It is used to predict how much work can be completed in future sprints and helps in accurate sprint planning.

Sprint-1 Velocity Metrics	Value
Story Points Planned	**14**
Story Points Completed	**14**
Sprint Duration	**7 Days**

Velocity (Points/Day)	**14 ÷ 7 = 2.0 story points/day**
Sprint-2 Estimation (12 pts)	**12 ÷ 2.0 = 6.0 days → Planned: 7 Days**

Use of Velocity:

- To estimate future sprints more accurately based on past performance.
- Sprint-2 has 12 story points. Based on velocity of 2.0 points/day: 12 ÷ 2.0 = 6.0 days → planned as 7 days.
- Velocity helps identify team capacity and flag if a sprint is overloaded before it begins.

Sprint-1 Burndown Chart Data

The burndown chart tracks the remaining work (story points) against elapsed time across Sprint-1. An ideal burndown line shows steady reduction from 14 points on Day 1 to 0 on Day 7. The table below shows the actual vs ideal burndown for Sprint-1.

Day / Date	Ideal Remaining Points	Actual Remaining Points	Points Completed (Day)	Cumulative Completed	Status
Day 1 (4 Mar)	14	12	2	2	**On Track**
Day 2 (5 Mar)	12	9	3	5	**On Track**
Day 3 (6 Mar)	9	7	2	7	**On Track**
Day 4 (7 Mar)	7	4	3	10	**On Track**
Day 5 (8 Mar)	4	2	2	12	**On Track**
Day 6 (9 Mar)	2	0	2	14	**Completed**
Day 7 (10 Mar)	0	0	0	14	**Sprint Done**

Burndown Chart Interpretation:

- Sprint-1 started with 14 story points on 4 March 2026 and was completed by 10 March 2026 with all 14 points delivered.
- The team maintained a steady pace of 2 story points per day throughout Sprint-1 with no blockers or delays.
- The actual burndown line tracked closely with the ideal burndown line, indicating good sprint planning and execution.
- Sprint-1 successfully delivered: MySQL setup, Gender vs Heart Disease chart, Age vs Heart Disease chart, and Smoking & Alcohol impact visualization.

Story Points & Sprint Summary

Sprint	Total Stories	Total Story Points	Duration	Status
Sprint-1	4 Stories (USN-1 to USN-4)	14 Points	7 Days	Completed
Sprint-2	6 Stories (USN-5 to USN-10)	12 Points	7 Days	In Progress
TOTAL	**10 Stories**	**26 Points**	**14 Days**	**—**

Definition of Done (DoD)

A User Story is considered DONE when all of the following criteria are met:

- Tableau visualization is created and matches the user story acceptance criteria.
- All filters (age, gender, race, lifestyle) work correctly on the visualization.
- Data values are verified against the MySQL database — no discrepancies.
- Chart labels, titles, and axis names are correctly formatted and readable.
- The visualization is added to the Tableau Dashboard and Story layout.
- The workbook is saved as a packaged .twbx file with data included.
- Dashboard is published to Tableau Public and accessible via public URL.